

Lab 1 – Ideal Gas Law

CHE 260 – Thermodynamics and Heat Transfer

Group Members

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1 Introduction

2 Results and Discussion

2.1 Part 1: Path Independence and Tank Volume Ratio

Part 1 of the lab discusses the comparison between the movement of air between two identical volume tanks, one with a gauge pressure of 40 PSI to another with a gauge pressure of 0-PSI. The main point of comparison is the differences in a slow equalization of pressure versus a rapid equalization of pressure.

2.1.1 Processed Results: Rapid and Slow Expansion

Figures 1–6 show the measured pressure, temperature, and mass-flow histories for the rapid and slow expansions.

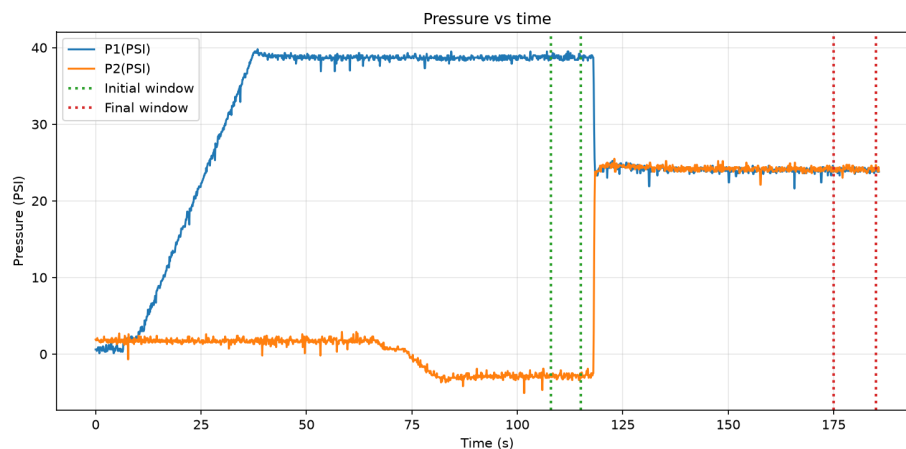


Figure 1: Tank pressures during the Part 1a rapid expansion.

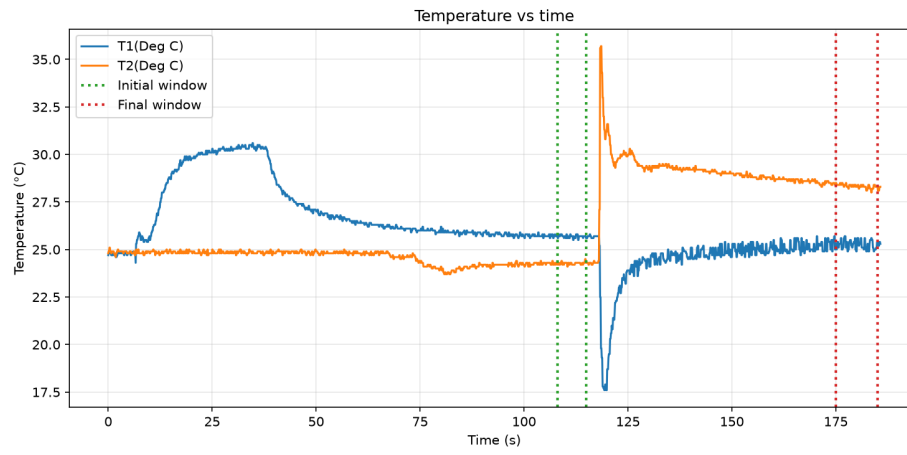


Figure 2: Tank temperatures during the Part 1a rapid expansion.

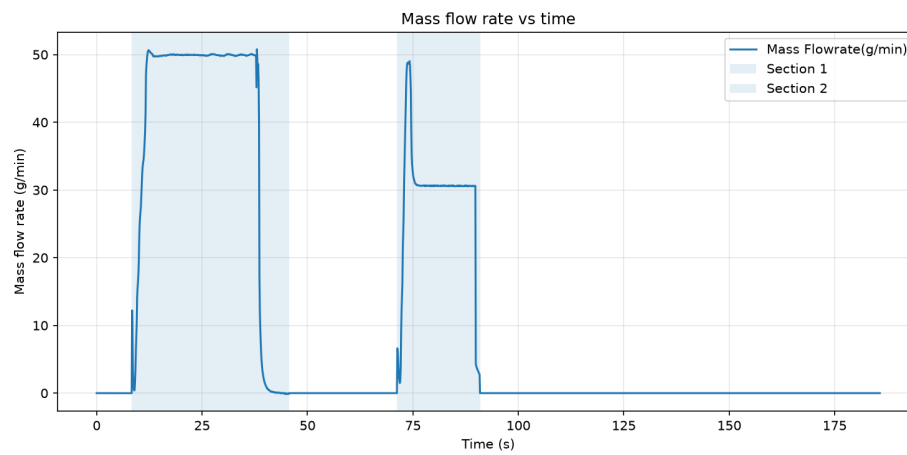


Figure 3: Mass flow rate during the Part 1a rapid expansion.

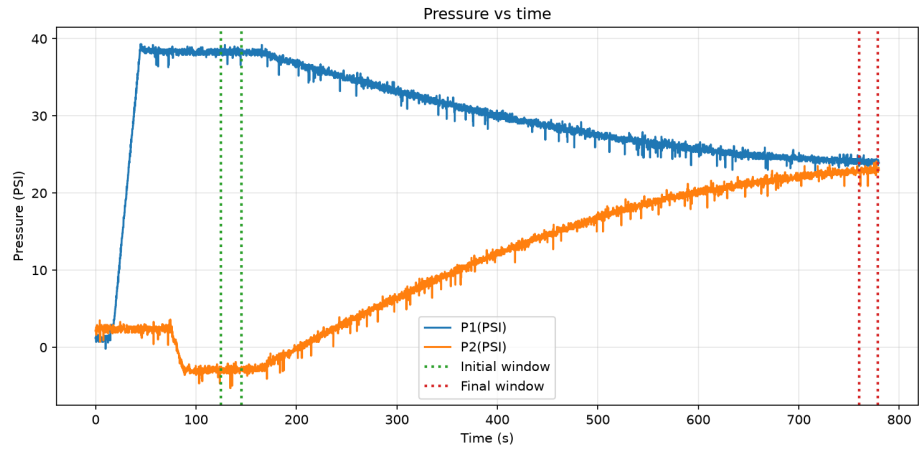


Figure 4: Tank pressures during the Part 1b slow expansion.

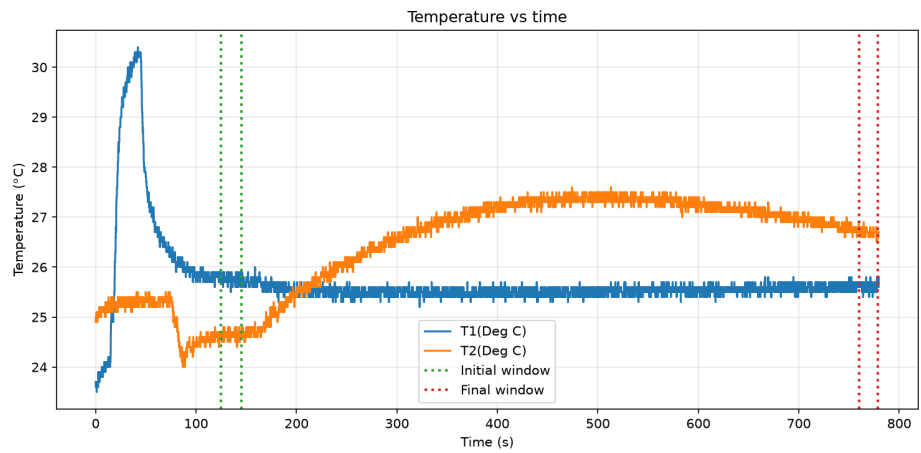


Figure 5: Tank temperatures during the Part 1b slow expansion.

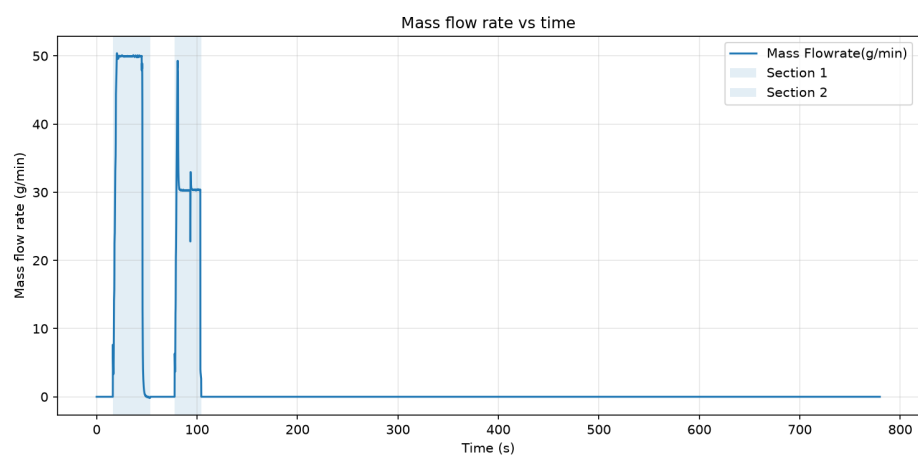


Figure 6: Mass flow rate during the Part 1b slow expansion.

2.1.2 Tank Volume Ratio

2.1.3 Heat Transfer and Net Heat Transfer

2.1.4 Path Independence of State Properties

2.2 Part 2: Initial Mass and Volume of the Left Tank

2.2.1 Processed Pressure, Temperature, and Mass Flow Results

2.2.2 Initial Mass and Tank Volume

2.2.3 Sources of Error and Limitations

2.2.4 Effect of Air Compressibility

The compressibility of air can greatly affect experimental results of pressure and volume, especially as experimental conditions diverge from assumptions made by the ideal gas equation. The ideal gas equation assumes that temperatures are kept high and pressures are kept low. When pressures are kept low, intermolecular forces can be assumed to be negligible. With this model, the compressibility of air can be modelled to be constant. As the pressure increases, electromagnetic repulsions and attractions between particles greatly contribute to the behavior of the gas. This causes the linear model of air compressibility to break down, displaying the inaccuracies in the ideal gas model. Since the intermolecular forces increase as pressure increases, our calculated compressibility is much higher than the actual compressibility.

One can model this using the compressibility factor, where

$$Z = \frac{Pv}{RT}$$

or

$$PV = ZRT.$$

Under constant volume, this measures the divergence of a physical model from Gay-Lussac's Law.

During our experimentation, the maximum gauge pressure found was around 40 psi. Given that this is a similar gauge pressure to that seen in pressurized car tire, this is high enough where our apparatus diverges from assumptions made by the ideal gas model and we should check the compressibility factor during our process.

If we hold the volume V constant, we can calculate the compressibility factors at low and high pressures.

At $t = 10$:

$$1.3 \cdot v = Z_1 \cdot R \cdot 317.85,$$
$$Z_1 = \frac{1.3 \cdot v}{317.85 \cdot R} = \frac{2}{489} \cdot \frac{v}{R}.$$

At $t = 50$:

$$52.2 \cdot v = Z_2 \cdot R \cdot 301.45,$$
$$Z_1 = \frac{52.2 \cdot v}{301.45 \cdot R} = \frac{1050}{6029} \cdot \frac{v}{R}.$$

Thus, through these calculations, we can see that the compressibility of air significantly increases at higher pressures.

3 Conclusion

References